

**PURCHASE DESCRIPTION FOR THE ACQUISITION OF TWO
(2) HARD BEARING DUAL DRIVE HORIZONTAL BALANCING
MACHINES**

1.0 SCOPE.

1.1 Scope. This specification defines the minimum requirements for two (2) hard bearing, dual-drive, horizontal balancing machines. Both machines must be fully capable of being calibrated and certified to SAE AS8617 standards, including all associated reporting. The two machines shall be designated as follows: Machine 1 shall meet or exceed the specifications for ARP 4048, Class 1000. Machine 2 shall meet or exceed the specifications for ARP 4048, Class 3000. The contractor is responsible for the complete installation, qualification, and certification testing of the equipment at the final location: Building 3001, Tinker AFB / Oklahoma City-Air Logistics Complex (OC-ALC). The intended purpose of the machines is for balancing aerospace jet engine rotors for the 76th Propulsion Maintenance Group (PMXG).

2.0 APPLICABLE DOCUMENTS. The documents listed in this section are cited in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed in section 2. Unless otherwise specified, the documents listed in the solicitation or contract shall be the current edition at time of solicitation.

2.1 Government Documents.

2.1.1 Specifications, Standards, and Handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein.

STANDARDS – FEDERAL

FED-STD-H28B: Screw Thread Standards for Federal Service

FED-STD-313F: Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished to Government Activities

**U.S Department of Labor, Occupational Safety and Health Administration
(OSHA)**

29 CFR 1910: General Industry, OSHA Safety and Health Standards

(Application for copies should be addressed to the OSHA Publications
Superintendent of Documents P.O. Box 37535 Washington, D.C. 20013-7535)

Environmental Protection Agency (EPA)

40 CFR 82: Protection of Stratospheric Ozone

(Application for copies should be addressed to the EPA Ariel Rios Building
1200 Pennsylvania Avenue, N.W. Washington, DC 20460)

MILITARY STANDARDS

MIL-STD-130N w/Change 1: Identification Marking of U.S. Military Property
Item Unique Identification Marking (IUID)

(Application for copies should be addressed to Air Force Product Data Systems
Modernization Standards (TMSS) Office, 554 ELSG/SBP, 4375 Chidlaw Road,
Bldg. 262, Room S008, Wright-Patterson AFB, OH 45433-5006, Phone (937) 257-
0870 or 0853 or <https://quicksearch.dla.mil/qsSearch.aspx>)

2.1.2 Other Government Documents, Drawings, and Publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the documents listed in the solicitation or contract shall be the current edition at time of solicitation.

LOCAL FACILITY DOCUMENTS

LOCAL ATTACHMENTS

Attachment A: H6WP40 - Floor Plan Drawing

2.2 Non-Government Publications, Specifications, Standards, and Handbooks. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the documents listed in the solicitation or contract shall be the current edition at time of solicitation.

American Gear Manufacturers Association (AGMA)

AGMA ISO 1328-1-B14: Cylindrical Gears - ISO System of Flank Tolerance
Classification - Part 1: Definitions and Allowable Values of Deviations Relevant to
Flanks of Gear Teeth

(Address application for copies to: 500 Montgomery Street, Suite 350, Alexandria,
VA 22314-1581 USA)

American National Standards Institute, Inc. (ANSI)

B11/STD B11: General Safety Requirements Common to ANSI B11 Machines
S2.60: Balancing Machines - Enclosures and Other Safety Measures

(Address application for copies to the American National Standards Institute, 11
West 42nd Street, New York, NY 10036)

International Organization for Standardization (ISO)

ISO 21940-23:2012: Mechanical vibration: Rotor balancing

ISO 4413: Hydraulic fluid power-General rules and safety requirements for systems and their components

ISO 4414: Pneumatic Power-General rules and safety requirements for systems and their components

ISO 7475: Enclosures and Protective Measures

(Application for copies should be addressed to the International Organization for Standardization (ISO) 1 rue de Varembe, Case postale 56, CH-2111 Geneva 20, Switzerland)

National Electrical Manufacturers Association (NEMA)

NEMA 250: Enclosures for Electrical Equipment (1000 V Maximum)

NEMA MG-1: Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 1300 N 17th Street, Rosslyn, VA 22209)

National Fire Protection Association (NFPA)

NFPA 70: National Electrical Code

NFPA 79: Electrical Standard for Industrial Machinery

(Applications for copies should be addressed to the National Fire Protection Association, ATTN: Support Services, 11 Tracy Drive, Avon, MA 02322)

Society of Automotive Engineers (SAE)

AS8617: Balancing Machines Verification Test Requirements

ARP4048: Balancing Machines- Description and Evaluation Horizontal, Two-Plane, Hard-Bearing type for Gas Turbine Rotors

ARP4162: Balancing Machine Proving Rotors

(Application for copies should be addressed to: Society of Automotive Engineers 400 Commonwealth Drive, Warrendale PA 15096)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulation unless a specific exemption has been obtained.

2.4 Acronym Definitions Not Explicitly Listed in Text. The following acronyms not explicitly listed in each section text are defined as follows:

- a. JIS- Japanese Industrial Standards
- b. DIN- German Institute for Standardization
- c. EN- European Standard

2.5 Numerical Definitions. The government will follow the following mathematical conventions:

- a. Whole numbers will be expressed where one thousand is written as 1,000.
- b. Decimals will be expressed where one-tenth is written as 0.1.

3.0 REQUIREMENTS.

3.1 Design. The item offered shall be new and a current commercially available model. A current model is defined as the manufacturer's currently produced model which, on the date this solicitation is issued, has been designed, engineered, sold or is being offered for sale through advertisements or manufacturer's published catalogue or brochures. Products such as a prototype unit, pre-production model, or experimental unit do not qualify as meeting the requirements specified herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage or distortion shall be accessible for adjustment, replacement, and repair.

3.1.1 Measuring and Indicating Device Graduations. All measuring and indicating devices on the machine shall be graduated in the United States customary units. An electronic control system that demonstrates switching between manufacturer standard units to United States customary units will be found acceptable.

3.1.2 Controls. All electrical, mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's workstation(s).

3.1.3 Safety and Health Requirements. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910 or equivalent ISO/EN/DIN/JIS standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

3.1.3.1 Safety Guarding. Covers, guards, or other safety devices shall be provided per ANSI B11 or equivalent ISO/EN/DIN/JIS standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements. The safety devices shall not interfere with the operation of the machine. The safety devices

shall prevent unintentional contact with the guarded part, and shall be easily removed to facilitate inspection, maintenance, and repair of the parts.

3.1.4 Audible Noise Level. Audible noise emitted by the equipment shall not exceed 84 decibels (dB), measured on the “A” weighted scale of a standard Type II sound level meter, at the operator's work position or any point at a distance of three (3) feet from the equipment. Noise generated by the work piece shall be excluded in determining compliance of the equipment with the 84 dB requirements.

- a. Each of the machines shall have permanent noise abatement that reduces any internal noise by 20 decibels for noise created between the work supports.
- b. The safety/noise abatement enclosure shall conform to manufacturing ISO 21940-23 standards with protection class of C600

3.1.5 Hazardous Material Exclusions. Supplies or materials being provided as part of the equipment shall be free of known hazardous materials. These materials shall not be brought on site without prior approval from the Government. Radioactive materials or instruments capable of producing ionizing radiation as well as materials which contain asbestos, mercury, lithium, methylene chloride, lead (= or >0.06%), or polychlorinated biphenyls (PCB's) are prohibited. Definitions of hazardous materials are specified in the latest version, including revisions adopted during the term of the contract, of FED-STD-313F. Nickel Metal Hydride and lithium batteries are permissible for memory backup. Class I Ozone Depleting Substances as defined in 40 CFR 82 shall not be used in the performance of this contract or be provided as part of the equipment. Exceptions to the prohibition of these materials must be referred to the Contracting Officer in writing, for consideration, not later than 60 calendar days after effective date of contract and prior to any work being performed.

3.1.5.1 PCB Certification. Written certification from the manufacturer that the equipment contains no detectable PCBs (less than two (2) parts-per-million) shall be provided. The certification shall be on manufacturer's letterhead, and signed by a company official who is empowered to provide the same.

3.1.6 Environmental Protection. Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.1.7 Lubrication. Means shall be provided to ensure proper lubrication for all moving parts in accordance with manufacturer requirements. All oil holes, grease fittings, and filler caps shall be easily accessible.

3.1.7.1 Lubrication Data. The Contractor shall provide the following information on all liquid or grease lubricants contained within equipment/system or in bulk (5 gallons or more) supplied under this procurement.

- a. Liquid Lubricants
 - i. Viscosity grade in ISO units
 - ii. Viscosity Index

- iii. AGMA and/or SAE classification as applicable
- iv. Mil Spec and/or NATO Code as applicable
- v. Viscosity in Centistokes (cst) @ 40° C and 100° C
- vi. Type Additive package. Example: ZDDP, Rust inhibitors, Foam inhibitors.
- vii. Neutralization, the TBN (alkalinity) and/or TAN (acidity).
- viii. Flash point
- ix. Pour point
- b. Grease Lubricants
 - i. National Lubrication and Grease Institute (NLGI) Number
 - ii. Type and percent of thickener
 - iii. Dropping point
 - iv. Maximum Service Temperature
 - v. Base oil viscosity range in SUS or centipoise

3.1.7.2 Recirculating Lubrication. When the machine is equipped with a recirculating lubrication system, it shall include a filter that is cleanable or replaceable. Each lubricant reservoir shall have at least a 24-hour capacity. Means shall be provided to indicate a low lubrication condition. When a low lubrication condition occurs, the machine shall complete the current operation and come to a complete stop. The machine shall indicate that lubrication is required.

3.1.8 Replacement Parts. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts shall be interchangeable without requiring modification. Replacement parts are those that are subject to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include but not be limited to, motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Foreign-sourced replacement parts shall either be interchangeable with domestically supplied components or be readily available from a domestic source.

3.2 Construction. The machine shall be constructed of parts that are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without distortion.

3.2.1 Castings and Forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, filling with solder or paste shall be used for reclaiming any defective part.

3.2.2 Fastening Devices. All screws, pins, bolts, and other fasteners shall be installed to prevent unintentional loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.2.3 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28B and the applicable “Detailed Standard” section referenced therein or equivalent ISO/EN/DIN/JIS standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

3.2.4 Hydraulic System. Under the condition that a machine has a hydraulic system, the system shall conform to ISO 4413. The system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls.

3.2.5 Pneumatic System. Under the condition that a machine has a pneumatic system, the system shall conform to ISO 4414. The system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls.

3.2.6 Gears. All gears of axis drive trains shall conform to or exceed all provisions of ANSI/AGMA ISO 1328-1-B14. All gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in drive trains shall be hardened and ground steel.

3.2.7 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. The condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.2.8 Welding, Brazing, or Soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.2.9 Painting. The machine shall be painted in accordance with the manufacturer's commercial practice. All paint shall be of the lead-free type.

3.2.10 Workmanship. Workmanship of the machine shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this purchase description.

3.3 Components. The machine shall consist of not less than the following components:

3.3.1 Base. The base shall be a stress relieved, heavy duty, ribbed casting or steel weldment having characteristics that minimize both thermal and vibration effects, while providing the strength and size to assure maximum rigidity and support for any load within the rated capacity of the machine. Means shall be provided for accurate, ease of leveling.

3.3.2 Bed. Each balancing machine shall, at a minimum, provide for the rotor envelope stipulated in ARP 4048 for its respective class. Furthermore, all machines are required to possess the following specifications:

3.3.2.1. Machine 1 Bed. Bed shall have a minimum of 10 feet of axial movement for adjusting the work supports/pedestals.

3.3.2.2. Machine 2 Bed. Machine 2 bed shall have a minimum of 11 feet of axial movement for adjusting the work supports/pedestals/clamshell part shroud. Machine 2 bed shall have adequate clearance to balance a rotor of 73-inch length, using 500 ft-lb

drive shaft on closest two pedestals, while 85-inch diameter clamshell shroud and furthest pedestal are positioned/stored at opposite end of machine bed.

3.3.3 Pedestals. The machines shall be delivered with work support pedestals. To ensure operational flexibility, the pedestal closest to the end drive motor shall be axially positionable to allow for end-drive balancing configurations. The system shall also provide sufficient clearance for inboard twin roller bearing rotor supports, maintaining an adequate operational distance for a 53-inch center-to-center dimension.

3.3.3.1. Machine 1 Pedestal. Machine 1 shall have two (2) pedestals with a minimum of 10 feet of axial movement for adjusting the work supports/pedestals.

3.3.3.2. Machine 2 Pedestal. Machine 2 shall have three (3) pedestals. The pedestals shall have adequate clearance between pedestals to balance a rotor of 73-inch length, using 500 ft-lb drive shaft on the closest two pedestals, while 85-inch diameter clamshell shroud and furthest pedestals are positioned/stored at opposite end of machine bed.

3.3.4 Drive Shaft. Each of the balance machines shall be supplied with two (2) drive shafts. Each machine drive shaft is required to possess the following specifications:

3.3.4.1. Machine 1 Drive Shafts. Machine 1 shall be supplied with one (1) ARP 1384 60 ft-lb drive shaft and one (1) ARP 1384 300 ft-lb drive shaft.

3.3.4.2. Machine 2 Drive Shafts. Machine 2 shall be supplied with one (1) ARP 1384 500 ft-lb driveshaft and one (1) ARP 1384, 1000 ft-lb drive shaft.

3.3.5 End Drive Coupling. To accommodate the installation and indexing of splined rotor adapter balance tooling, the balance machine's end drive motors and couplings must provide both coarse and fine axial adjustment. The coarse adjustment mechanism is required to allow for at least 10 inches of axial movement.

3.3.6 Enclosure. Each balancing machine shall be contained within a C600-class safety and noise abatement enclosure conforming to ISO 7475 and ANSI S2.60 standards. The system must include permanent guards over all rotating components and feature safety interlocks that prevent operation if any safety device is not in place. The enclosure shall be electrically interlocked with the machine, with a supervisor key switch override for maintenance located inside the electrical cabinet. For access, the enclosure will have a primary, power-operated door that allows full overhead and frontal access to the machine bed, as well as a separate personnel door. Viewing windows shall be integrated into the front walls. The interior of the enclosure must be equipped with switch-operated working lights and an additional measurement/computer display. All electrical cabling shall be run through conduit or flexible trays. To ensure signal integrity, cables of different types must be run separately or shielded appropriately to prevent interference.

3.3.6.1. Machine 2 Enclosure Length. The total length of Machine 2 and its enclosure must not exceed 23 feet when measured parallel to the machine bed. This maximum dimension applies when the machine is in its fully extended operational state and represents the required footprint, not including any additional clearance needed for opening doors or panels.

3.3.7 Machine 2 Shroud. A Machine 2 compatible, clamshell-type part shroud shall be provided. It will feature an 85-inch diameter by 30-inch-long envelope and a 9.5-inch diameter concentric arbor opening. The shroud's opening and closing will be remotely actuated from the console and will include a safety interlock that prevents operation unless fully closed. For setup and storage, the shroud is designed to be moved axially along the bed and is fully removable using integrated lifting points and quick-disconnect fittings. The design must provide clearance for support pedestals at a 53-inch center-to-center dimension and allow full overhead access for rotor loading.

3.3.8 Balance System Control Computer. The balancing machine's control computer shall operate on the Microsoft Windows 11 LTSC operating system and be provided with a laser printer for reporting functions. The balancing software must support indefinite data storage for each run, both locally and to a networked location. Key software features shall include a tool compensation module, the ability to electronically zero a run for check tests, and password-protected user profiles that allow an administrator to restrict access to calibration functions without the aid of the OEM. All balance reports, Shall save as a PDF, are required to display a minimum of six lines for operator input, setup tolerances, tool compensation data, a rotor setup diagram, and a timestamp for each run. Furthermore, the report must clearly detail the unbalance amount and angle for each plane, the correction mode (add/remove), and indicate if tool compensation was active and if the run was within tolerance. The contractor must utilize Energy Star or EPA-accepted materials where applicable. Finally, all electrical components must be UL compliant.

3.3.9 Third Party Software License. The Contractor shall provide all necessary license(s) for third party applications used to meet the requirements outline in this statement of work. The Contractor shall transfer ownership of all licenses to the Government for sustainment over the lifecycle of the system. All license(s) provided to Government shall not be subscription based and shall not require a connection to the internet for its activation, use, or updating.

3.3.10 Network Connectivity and Data Rights.

3.3.11.1. Third Party Software License. The Contractor shall provide and transfer ownership of all necessary third-party software licenses to the Government. These licenses are required for the long-term sustainment of the system and must be perpetual and non-subscription based. Furthermore, all provided software must be fully functional in an offline environment, meaning an internet connection shall not be required for its activation, use, or updating.

3.3.11.2. Data Communication Protocols. The machine controller(s) shall be provisioned and licensed to communicate using industry-standard protocols, with OPC-

UA as a mandatory minimum. All OPC-UA Tag information must be made available as output for data collection by the Government's external industrial network. For physical connectivity, the controller shall be equipped with at least one RJ45 Ethernet port, configured for this purpose. This network port must be placed in an accessible location on the equipment that does not require major disassembly for access.

3.3.11.3. Account Privileges. The machine controller(s) and HMIs shall be configured with distinct user and administrative account privileges. User accounts are to be role-based, with permissions defined by job function (e.g., operator or maintenance). Critically, the system must be designed so that all user-level machine functions can be performed without requiring Windows administrative privileges, ensuring operational permissions are managed within the application layer.

3.3.11.4. Software Patches. Controller and peripheral equipment shall come with the latest OEM software version and patches installed at the time of final acceptance.

3.3.11.5. Security Vulnerabilities. At the time of destination acceptance, each machine shall comply with all published Security Advisories relating to the remediation of known security vulnerabilities associated with the installed control system(s), sub-system(s), and software. Workarounds and mitigation procedures shall not be the primary means of vulnerability remediation.

3.3.11.6. Wi-Fi Devices. Any Wi-Fi or similar wireless technology integrated into the equipment or its sub-components shall be FIPS 140-3 compliant (or the most recent edition). All wireless connections must use certificate-based authentication over the current 802.11 standard. The use of wireless technology shall not be the sole method of connecting the machine to the Government's industrial network; a primary wired connection must also be provided. The Contractor is required to declare all use of wireless technology to the Government POC no later than 30 days after the contract award. Furthermore, the Contractor is responsible for furnishing the Government with all necessary documentation to formally demonstrate this compliance.

3.3.11.7. Bluetooth®. The use of Bluetooth® technology on equipment or sub-component of that equipment is prohibited.

3.3.11 Uninterruptible Power Supply (UPS). Machine controllers shall be equipped with an uninterruptible power supply to safeguard installed industrial computer system(s) from damage in the event of main power interruptions. The alternative power supply shall be provisioned to automatically sense partial or total loss of main power and initiate a complete systematic shutdown the industrial computer(s).

3.3.12 Electrical System. The machines shall conform to NFPA 79 or current equivalent ISO/EN/DIN/JIS/IEC standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements. The existing source power

available at the facility is 480 volt, 3-phase, 60 Hertz (Hz) . The machine's electrical system shall be tolerant of fluctuation of ± 10 percent. The electrical system shall be complete, including any electrical transformer(s) that may be required to modify the existing source voltage to the proper operating voltage of the equipment. A properly rated, and fused, single disconnect device shall be utilized on the machine with means of lockout in the off position only. All Electrical components, wires, controls, drives, motors, sensors, relays, and any other applicable components shall be UL compliant.

3.3.12.1 Electrical Enclosures. Construction of the electrical enclosures shall be in accordance with NEMA 250 type 12 or current equivalent ISO/EN/DIN/JIS/IEC standards and shall provide for ventilation of components and shall be designed for an indoor non-hazardous location. All electrical components shall conform to NFPA 79 or equivalent ISO/EN/DIN/JIS standards. The machine shall be provided with interlocks on its electronic cabinet that shall instantly remove power from the equipment when the door is opened in accordance with NFPA 79 or current equivalent ISO/EN/DIN/JIS/IEC standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

3.3.12.2 Arc Flash Protection. The machine enclosures shall be marked for arc flash in accordance with NFPA 79 Paragraph 16.2.3 "Safety Signs for Electrical Enclosures" or current equivalent ISO/EN/DIN/JIS/IEC standards for arc flash safety. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

3.3.12.3 Solid-State Components. Electrical devices used to power electrical servo and spindle motors of control voltage delivered to solid-state electrical components shall be solid-state devices. Such devices shall be equipped to handle the electrical load imposed by operation of the machine to its maximum potential. The use of selenium in solid-state components shall be restricted to those devices that protect other solid-state components from voltage surge.

3.3.13 Motors. Motors shall be rated for continuous duty. Alternating-current (AC) motors shall be designed to operate on 60-Hz and each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall meet the requirements of NEMA MG-1 or current equivalent ISO/EN/DIN/JIS/IEC standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements. Each motor shall have an identification plate in accordance with paragraph 3.7 to identify manufacturer, model number, serial number, voltage, amperage, horsepower, phase, frequency, and frame number. Motor horsepower requirements shall be as stated in Table 1.

3.3.14 Work light. A work light, whether built-in or added on, shall be provided. The light shall have a protective shield and provide maximum light to the work area.

3.3.15 Facility Connections. Electrical connections made to the facility electrical system shall conform to NFPA 70.

3.3.16 Hour Meter. The machine shall be equipped with a non-resetting type, hour meter to display accumulated operating time of the drive motor(s), mounted on the main electrical panel visible from the outside, sealed to prevent the entrance of dust and moisture, and shall be mounted to withstand shock and vibration generated by the machine. The meter shall have a range of 0 to 99,999.9 hours in increments of one (1) tenth of one (1) hour. A CNC control that tracks and displays total machine on time (a non-resetting type), controller on, program runtime, and run time of the servo motor(s) shall be considered to meet this requirement. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.4 Size and Capacity. The size and capacity of the machine shall be not less than specified in Table I.

TABLE I
Machine 1

Total Load Capacity	4,500 pounds
Minimum Belt Drive Horsepower Rating	50HP/37.4KW
Minimum End Drive Horsepower Rating	50HP/37.4KW
Minimum Supported Pedestal Load	1,800 Pounds @ 1150RPM
$W \cdot n^2$ value	$W \cdot n^2_{\max} \geq 2400 \cdot 10^6 \text{ lb} \cdot \text{rpm}^2$
Maximum weight of rotor, belt drive	10,000 pounds
Maximum weight of rotor, end drive	10,000 pounds
Maximum weight of rotor, belt drive @ 900 RPM	2,500 pounds
Maximum weight of rotor, end drive @ 900 RPM	2,500 pounds
Balancing speeds variable, belt drive	500-1500 RPM
Balancing speeds variable, end drive	500-1500 RPM
Maximum length of rotor between bearing centers	106 inches
Maximum diameter of rotor over bed, end drive	69 inches in Diameter
Maximum diameter of rotor over belt tensioner, belt drive	50 inches in Diameter

Machine 2

Total Load Capacity	9,500 pounds
Minimum Belt Drive Horsepower Rating	50HP/37.4KW
Minimum End Drive Horsepower Rating	75HP/56KW
Minimum Supported Pedestal Load	3000 Pounds @ 1200 RPM
$W \cdot n^2$ value	$W \cdot n^2_{\max} \geq 5000 \cdot 10^6 \text{ lb} \cdot \text{rpm}^2$
Maximum weight of rotor	15,000 pounds
Maximum weight of rotor, @ 900 RPM	5,000 pounds
Balancing speeds variable, belt drive	500-1500 RPM
Balancing speeds variable, end drive	500-1500 RPM
Maximum length of rotor between bearing	106 inches

centers	
Maximum diameter of rotor over bed	69 inches in Diameter
Maximum diameter of rotor over belt tensioner, belt drive	50 inches in Diameter

3.5 Alignment and Accuracy Tolerances. The alignment and accuracy tolerances of the machine shall be as specified in Table II.

TABLE II
Machine 1

Accuracy (displacement of C.G.) Belt Drive	≤10 Micro inch
Accuracy (displacement of C.G.) End Drive	≤15 Micro inch
Unbalance reduction ratio	95%
Sensitivity, Belt drive	≤0.25 gmm/kg
Sensitivity, end drive	≤0.38 gmm/kg

Machine 2

Accuracy (displacement of C.G.) Belt Drive	≤10 Micro inch
Accuracy (displacement of C.G.) End Drive	≤15 Micro inch
Unbalance reduction ratio	95%
Sensitivity, Belt drive	≤0.25 gmm/kg
Sensitivity, end drive	≤0.38 gmm/kg

3.6 Standard Tooling and Equipment. All standard tooling and equipment required for proper operation and maintenance of the machine shall be provided. A list of these items shall be furnished at time of proposal.

3.6.1. Required Tooling: The following tooling shall be supplied with the machines:

- a. The Contractor shall provide two (2) ARP4162 horizontal-proven rotors, along with appropriate manufacturing and certification documentation for each proving rotor. One (1) of the two proving rotors shall be applicable for Machine 1. One (1) of the two proving rotors shall be applicable for Machine 2.
- b. Each of the balance machines shall be supplied with two pairs of SAE 13-inch pedestal insert saddles.
- c. Each of the balance machines shall be supplied with two pairs of pedestal roller assemblies that are capable of running with shaft diameters of 1.5 in – 7.00 inch.
- d. Each of the balance machines shall be supplied with two drive belts to spin the proving rotor.
- e. Each of the balance machines shall be supplied with two drive shafts to spin the proving rotor.

3.7 Marking on Instruments, Control Panels, and Plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion resistant.

3.7.1 Lubrication Plate. A lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information.

- Points of Lubricant Application
- Servicing Interval
- Type of Lubricant
- Viscosity
- Military or Federal Specification for Each Lubricant (If available)
- Size and Type of Each Lubricant Filter

3.7.2 Nameplate. A nameplate shall be permanently and securely attached to each machine. The nameplate data shall meet the requirements of NFPA 79 and contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine, and a suffix identified in the manufacturer's permanent records.

- Nomenclature
- CAGE Code
- Manufacturer's Name
- Manufacturer's Model Designation
- Manufacturer's Serial Number
- Power Input (Volts, Total Amps, Phase, Frequency)
- Contract Number
- Date of Manufacture
- Full-load current for each supply
- Ampere rating of the largest motor or load
- Short-circuit current rating of the industrial control panel
- Max ampere rating of short-circuit and ground fault protective device, where provided
- Electrical diagram number(s) or number of the index to the electrical drawings

3.7.3 Identification Marking of Military Property. An Item Unique Identification Marking (IUID) shall be provided in accordance with MIL-STD-130N and all applicable documents within the standard with Machine Readable Information (MRI) for item identification marking and automatic data capture. The application of Human Readable Information (HRI) shall be used in combination with MRI and free text. At a minimum this tag shall contain the information listed in paragraph 3.7.2 Nameplate.

3.8 Technical Data. Three (3) hard copies and one (1) electronic copy(s) of technical data shall be provided with the machine. Technical data shall contain specifications, instructions

for the operation, maintenance of the machine and recommended spare parts list. All copies of technical data shall be legible and written in the English language.

3.8.1 Plan of Action and Milestones (POA&M). The contractor shall provide a POA&M to identify contractor performance dates of all major events from contract award to completion of all contract requirements. The initial POA&M shall be delivered thirty (30) calendar days After Contract Award (ACA). The milestones shall be prepared using a recognized project management technique (preferably Microsoft Project). The Government and the contractor shall review the report on a monthly basis. The contractor shall notify the Government prior to making any changes to the POA&M. All changes to the POA&M shall be approved in advance by the Government. The POA&M shall detail such items as engineering, procurement of major hardware, assembly, operational debug, factory acceptance, shipment, installation, training, and final acceptance test.

3.8.2 Acceptance Test Plan (ATP). The Contractor shall prepare an Acceptance Test Plan (ATP). The ATP shall be a step-by-step written instruction on how to verify compliance during Contract Quality Assurance Inspection and the Final Inspection/Acceptance Test specified in Sections 4.4 and 4.5. The Contractor shall develop and provide dimensioned drawings of the proposed test part specified in Paragraph 4.7.3. The Manufacturer's standard testing procedures shall be included in the ATP. Any procedures necessary for compliance with Section 4, which are not covered by the Manufacturer's standard testing procedures, shall be identified and included in the ATP. The initial ATP shall be submitted thirty (30) calendar days ACA for Government review and approval. The accepted test procedures shall become part of this contract.

3.9. Installation. The contractor is responsible for the complete turnkey installation of one (1) SAE 1000 class or greater and one (1) SAE 3000 class or greater hard bearing horizontal balance machines at Tinker AFB. This includes providing all necessary labor, tools, equipment, materials, parts, manuals, and Personal Protective Equipment (PPE) for the shipping, rigging, installation, certification, and qualification. In support of this, Tinker AFB will provide a 480VAC electrical supply and a 100 PSI compressed air supply within 30 feet of one of the rear corners of each machine. Contractor shall be responsible for all security requirements to access Tinker AFB.

3.9.1 Foundation. The winning contractor is responsible for determining the project's foundation specifications. While the existing concrete foundation is a minimum of 8 inches thick, the contractor must subcontract any additional foundation evaluation beyond OC-ALC's thickness measurement capability. Should OEM requirements necessitate a thicker foundation, the contractor will bear the responsibility for its removal and the installation of a new one.

3.9.2 Lead and Asbestos Testing. Lead and Asbestos testing shall be performed by the 72nd ABW /CE prior to delivery by the contractor.

3.9.3 Work Stoppage. For unforeseen issues that are found during installation of the foundation, the contractor shall notify the facility point of contact (POC) and DLA immediately to determine further course of action.

3.10 Training. After satisfactory completion of acceptance testing of the system, the services of a qualified representative(s), who is proficient in the English language, shall be provided for specialized training to familiarize Government personnel with the equipment and to help ensure reliable performance and maximum service life, during normal usage. The training shall be for five (5) operators. Training shall be performed at the Government facility and shall not be less than five (5) consecutive eight (8) hour workdays, between 7:00 am and 3:30 pm excluding weekends and federal holidays. The course shall utilize as-built drawings and manuals provided under this contract and shall include both classroom and hands-on portions. All printed training aids shall be supplied for the course, be in the English language, and become the property of the Government.

4.0 QUALITY ASSURANCE.

4.1 Responsibility for Inspection. The contractor shall be responsible for the performance of all inspection requirements specified herein and in accordance with the attached Quality Assurance Provision (QAP). Except as otherwise specified in the contract or purchase order, the contractor may use his own facility for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to witness any inspections set forth in the purchase description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for Compliance. The Contractor shall be responsible for ensuring all items meet the requirements of sections 3 and 4.

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

- a. Contract quality assurance inspection (Origin) (see 4.4).
- b. Final Inspection/Acceptance test (Destination) (see 4.5).

4.3 Inspection Conditions. All inspections, tests, and examinations shall be performed in an indoor facility with ambient temperature conditions in the range of 55° F to 100° F and 10% to 95% relative humidity.

4.4 Contract Quality Assurance Inspection (Origin). Contract quality assurance inspection shall be applied at the contractor's facility prior to being offered for acceptance under the contract. Each machine and all peripheral equipment added to the machines shall be fully assembled and perform all functions in accordance with this specification. Quality conformance inspection shall consist of the examination in 4.6 and the tests in 4.7. The machines shall pass the examination, all tests, and contract quality assurance inspection to be acceptable for shipment.

4.5 Final Inspection/Acceptance Test (Destination). Final Inspection and acceptance test shall be performed on each machine to ensure conformance with this purchase description. The acceptance test shall be performed only after each machine is installed at its final

location. The acceptance test shall consist of the examination in 4.6 and all tests in 4.7. Each machine shall pass the examination and all tests to be accepted.

4.6 Examination. The equipment shall be examined for design, dimensions, construction, materials, components, electrical equipment and workmanship to determine compliance with the requirements of this specification in accordance with ARP4048.

4.7 Tests. Unless otherwise specified within this document, all programs, instruments, materials, material handling equipment, qualified personnel, and tools required to perform and evaluate the tests shall be furnished by the contractor. All measurement tools and indicating devices shall be calibrated in accordance with the attached QAP. If a malfunction occurs during the test, the test shall be restarted each time until all tests in 4.7 are completed without a malfunction. Three (3) consecutive failures without completion of the test may be considered cause for rejection of the system. For the purpose of this test, a “failure” is defined as any equipment malfunction which requires remedial action to restore the system to full operation in accordance with this specification.

4.7.1 Alignment and Accuracy Tests. Tests shall be conducted to ensure that each machine meets the alignment and accuracy tolerances specified in paragraph 3.5, Table II. Each machine’s repeatability and accuracy for tolerances specified in AS 8617 shall be checked.

4.7.2 Performance Test. Each machine shall be tested in accordance with the agreed upon ATP referenced in paragraph 3.8.2. Each machine shall be tested in accordance with AS8617.

4.7.3 Audible Noise Level Test. The equipment shall be tested for compliance with paragraph 3.1.4. “Audible Noise Level”.

INSTALLATION RESPONSIBILITIES

Government	Contractor	
☐	☒	a. Provide each machine a foundation in accordance with the Manufacturer's design requirements.
☐	☒	b. Furnish labor and material handling equipment for off-loading and placing item on foundation.
☐	☒	c. Provide and install anchor bolts and nuts.
☐	☒	d. Set and rough level each machine on its foundation
☐	☒	e. Level and align machines.
☐	☒	f. Connect each machine to the provided utilities hook-up.
☐	☒	g. Provide all necessary tools, gages, and instrumentation necessary to perform the required tests.
☐	☒	h. Provide and charge all systems with fluids in accordance with manufacturer's instructions.
☐	☒	i. Material for performance testing at Government facility
☐	☒	j. Material for performance testing at Contractor's facility
☐	☒	k. Material handling, Rigging equipment and Qualified Personnel for performance testing.

GOVERNMENT RESPONSIBILITIES

- ☒ Provide utilities hook-up within 30 feet, line of sight of each machine within the same compartment.
- ☒ No work shall be performed by customer personnel when the contractor's representative is not in the immediate work area during installation, verification, and initial start-up of each machine.